



Proposal

Kalam Innovation Research Academy (KIRA)

A Tech Incubator School from Grade 1



Vision

To create the **world's first Indian-origin "Research Incubator School"** where technology exposure starts from Grade 1, nurturing children into **innovators, researchers, and technopreneurs** by the time they graduate at 18. Inspired by Dr. A.P.J. Abdul Kalam's vision, KIRA will produce students who not only excel academically but also **file patents, publish papers, launch startups, and solve global challenges.**

Mission

- **Reimagine education:** Blend CBSE academics with 60–70% practical innovation labs, tech incubation, and research.
- **Produce creators, not job seekers:** Every graduate leaves with a portfolio of prototypes, research publications, and real-world projects.
- **Global responsibility:** Cultivate ethical innovators who apply technology for sustainability, peace, and inclusive growth.
- **Inclusivity:** Open 20–30% of seats for underprivileged/madarsa students with scholarships, bridging tradition with modern science.

How This School Is Different

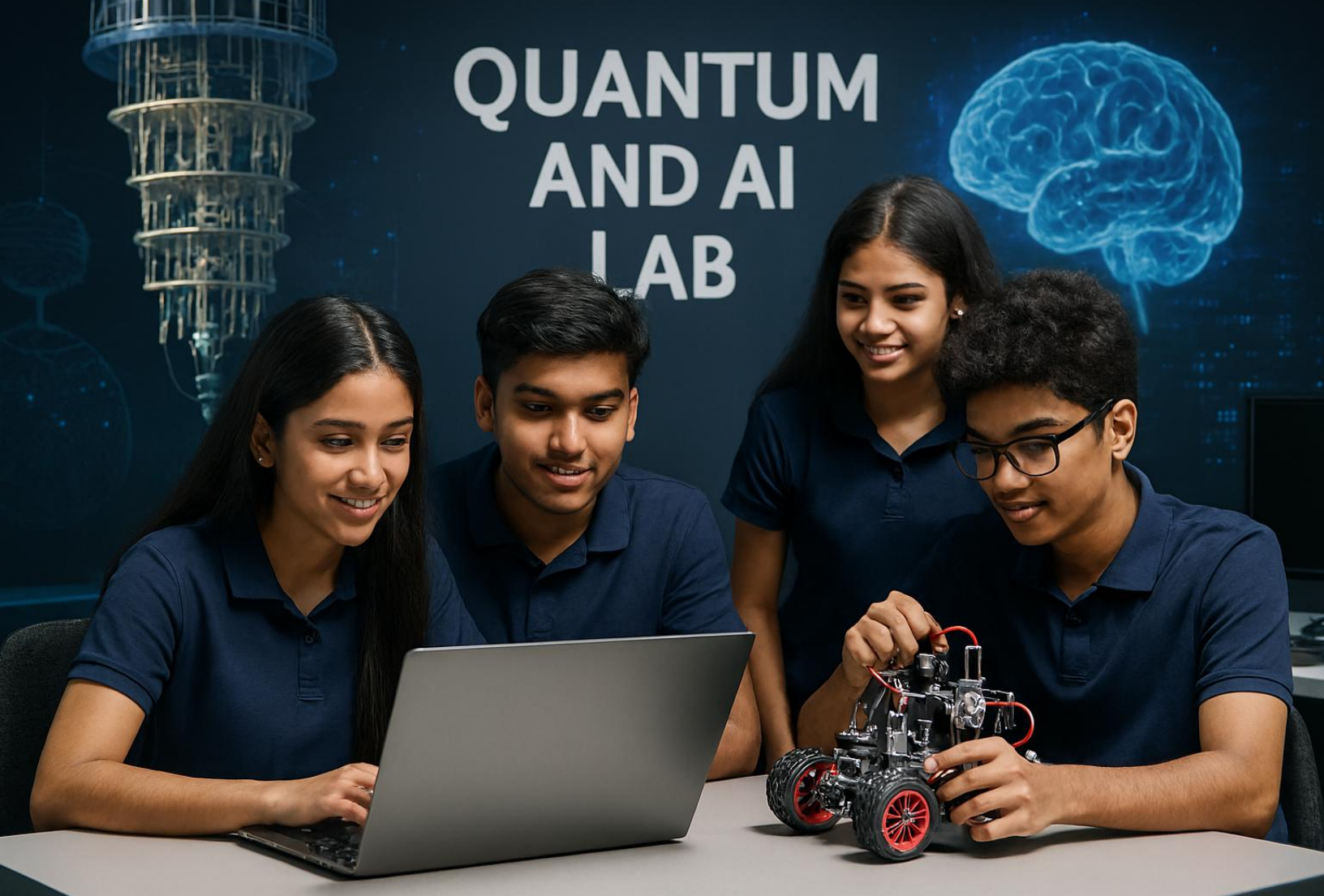
Compared to Indian Schools

- **Indian norm:** CBSE/ICSE focus on exams, rote learning, and coaching for IIT-JEE/NEET.
- **KIRA:** Focus on **innovation outcomes** – patents, startups, global projects. Assessment via portfolios, not exams.

Compared Globally

- **Examples:**
 - **High Tech High (USA)** – project-based learning.
 - **TKS – The Knowledge Society (Canada)** – prepares innovators in AI, biotech.
 - **Minerva University (USA)** – experimental liberal arts + innovation.
- **KIRA's uniqueness:** Begins **at age 6 (Grade 1)**, not at university/teen years. Students grow in a **living lab ecosystem** unmatched globally.

QUANTUM AND AI LAB



Academic + Tech Incubator Model

Grades 1–5: Early Curiosity Labs (Ages 6–10)

- **Academics:** NCERT basics taught via projects (fractions via coding a pizza game, science via dismantling gadgets).
 - **Daily Tech Hour:** Scratch, block coding, LEGO robotics, digital art.
 - **Creativity Labs:** Storytelling, design, maker crafts.
 - **Ethics & Values:** Simple lessons in digital citizenship and innovation for good.
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Grades 6–8: Foundation Research Fellows (Ages 11–13)

- **Academics:** Project-integrated math, science, and English.
- **Tinker Labs:** IoT sensor kits, Arduino basics, robotics prototypes.
- **Mini Research Projects:** Students publish short papers/journals.

- **Cultural Integration:** Islamic/Indian philosophy + ethics of science (for rooted global citizens).
 - **Exposure:** Visits to ISRO, IIT labs, local innovators.
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Grades 9–10: Innovation Apprentices (Ages 14–15)

- **Tech Specializations:** Python, AI basics, cybersecurity, web/mobile apps.
 - **Research Apprenticeships:** Tie-ups with IITs, DRDO, or industry labs.
 - **Innovation Challenges:** Solve SDG problems (e.g., water, energy).
 - **Failure Festivals:** Celebrate mistakes → reinforce experimental mindset.
 - **Global Collaborations:** Virtual MIT/Stanford exchange projects.
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Grades 11–12: Tech Leaders (Ages 16–18)

- **Capstone Research:** Year-long theses in areas like neuromorphic chips, quantum computing, AI for climate.
 - **Incubator Program:** Real startups, patent filings, industry mentorship.
 - **Global Exposure:** Present at TEDx Youth, international journals, coding competitions.
 - **Leadership & Mentorship:** Seniors guide juniors in labs (“Kalam Fellows”).
 - **Graduate Outcomes:** Every student leaves with:
 - At least one patent application
 - One published research paper
 - One startup prototype
 - Global project experience
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Daily Routine & Integration with CBSE

- **7 AM:** Wellness & techno-ethics (Yoga, digital responsibility).
- **8–12 PM:** CBSE Core subjects (integrated with projects).

- **1–3 PM:** Tech Labs (AI, robotics, coding).
- **4–6 PM:** Research, incubation, or clubs.
- **Evening:** Innovation clubs, open-source contributions.

Assessment System

- No rote exams.
 - **Portfolios, prototypes, publications, patents, impact scores.**
 - Peer review + quarterly “Innovation Defense” (like mini PhD viva).
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Global & National Impact

For India

- Breaks the cycle of rote-learning + unemployment.
- Creates innovators at 18 instead of graduates at 25.
- Bridges **madarsa + mainstream gap** through scholarships and feeder programs.

For the World

- India becomes a hub for **youth innovation exports**.
 - Global collaboration on solving climate, energy, health, and space challenges.
 - Establishes a rare **school-to-startup pipeline** unmatched by most countries.
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Why the World Needs This School (Need of the Era)

- **AI & Tech Disruption:** Jobs will vanish; innovators will thrive.
- **India’s Demographic Advantage:** Harness youth into creators, not clerks.
- **Global Leadership:** India can produce the next **Kalam, Musks, or Einsteins**—at school level.
- **Rare Concept:** Only scattered schools (TKS Canada, High Tech High USA) approach this model, but **none integrate CBSE + full-time research incubation starting from Class 1.**

FINANCIALS

Block-Wise Breakdown

Block	Sq Ft (from FTLA)	Civil/Structural Cost (₹/sq ft)	Civil Total (₹ Cr)	Interiors/Fixtures (₹ Cr)	Equipment/Fit-Outs (₹ Cr)	Block Total (₹ Cr)	Notes
Admin Section (Offices, Reception, IT Support)	15,000	₹2,500-3,500 (mid-range office-like)	0.38-0.53	0.15-0.20 (furniture, AC)	0.10-0.15 (computers, security systems)	0.63 - 0.88	Basic; scalable with biometric access.
Classrooms (30 Smart Classrooms + Study Areas)	50,000	₹2,000-3,000 (standard school)	1.00-1.50	0.40-0.60 (desks, boards)	0.50-0.75 (interactive displays, projectors @ ₹1-2 lakh/room x 30)	1.90 - 2.85	Includes flexible seating; smart tech via govt broadband schemes.
Libraries & Study Centers (Main Library, Digital Media)	20,000	₹2,500-3,500 (quiet zones)	0.50-0.70	0.20-0.30 (shelving, pods)	0.15-0.25 (digital resources, e-readers)	0.85 - 1.25	24/7 access with biometrics; books/digital at ₹50,000-

							1 lakh initial.
Labs (Tech + Science) (Coding, AI, Robotics, Physics/Chem/Bio)	330,000 (2.5L Tech + 0.8L Science)	₹3,000-4,500 (specialized: ventilation, safety)	9.90-14.85	3.30-4.95 (benches, wiring)	5.00-7.50 (high-end: GPUs @ ₹50k/unit x 100, robots @ ₹5-15 lakh/lab x 10-15 labs; ATL grants offset ₹20 lakh/lab)	18.20-27.30	Core for KIRA; phased equipping (e.g., ₹5-8 lakh basic STEM lab, scaled up).
Auditoriums & Event Spaces (Main 800-seat, Secondary Halls)	40,000	₹4,000-5,000 (acoustics, staging)	1.60-2.00	0.60-0.80 (seating, lighting)	0.80-1.20 (AV systems, 4K LEDs @ ₹10-20 lakh/setup)	3.00-4.00	For conferences; similar to 1000-seat projects at ₹5-10 Cr total.
Boys' Hostel (300 capacity, Junior/Senior)	150,000	₹2,800-3,500 (premium residential)	4.20-5.25	1.50-2.00 (beds, lockers)	0.80-1.20 (WiFi, security, lounges with VR/gaming)	6.50-8.45	Includes makerspaces; per bed ₹2-3 lakh total fit-out.
Girls' Hostel (300 capacity,	150,000	₹2,800-3,500 (same as boys)	4.20-5.25	1.50-2.00 (beds, wellness centers)	0.80-1.20 (same as boys, plus women-in-	6.50-8.45	Mirror design for equity; enhance

Junior/Senior)					tech suites)		d security.
Staff/Guest Residences (VIP Suites, Faculty Housing)	80,000	₹2,500-3,500 (residential)	2.00-2.80	0.80-1.00 (kitchens, furnishings)	0.40-0.60 (conference facilities, basic tech)	3.20 - 4.40	For 20-30 staff; attracts talent with premium amenities.
Other/Common Areas (Dining, Sports, Maintenance)	15,000 (estimated residual)	₹2,000-3,000	0.30-0.45	0.15-0.20	0.10-0.15 (kitchen equip, sports gear)	0.55 - 0.80	Shared facilities; often 10% of total area.

Grand Total (Before Add-Ons): ₹41.33-58.38 Cr

Conclusion

KIRA is not just a school — it is a movement.

It will redefine how the world views Indian education: producing teenagers with global patents, startups, and research — not job applications.

“Where a 12-year-old builds robots, a 15-year-old publishes AI research, and an 18-year-old launches a startup — all before college.”